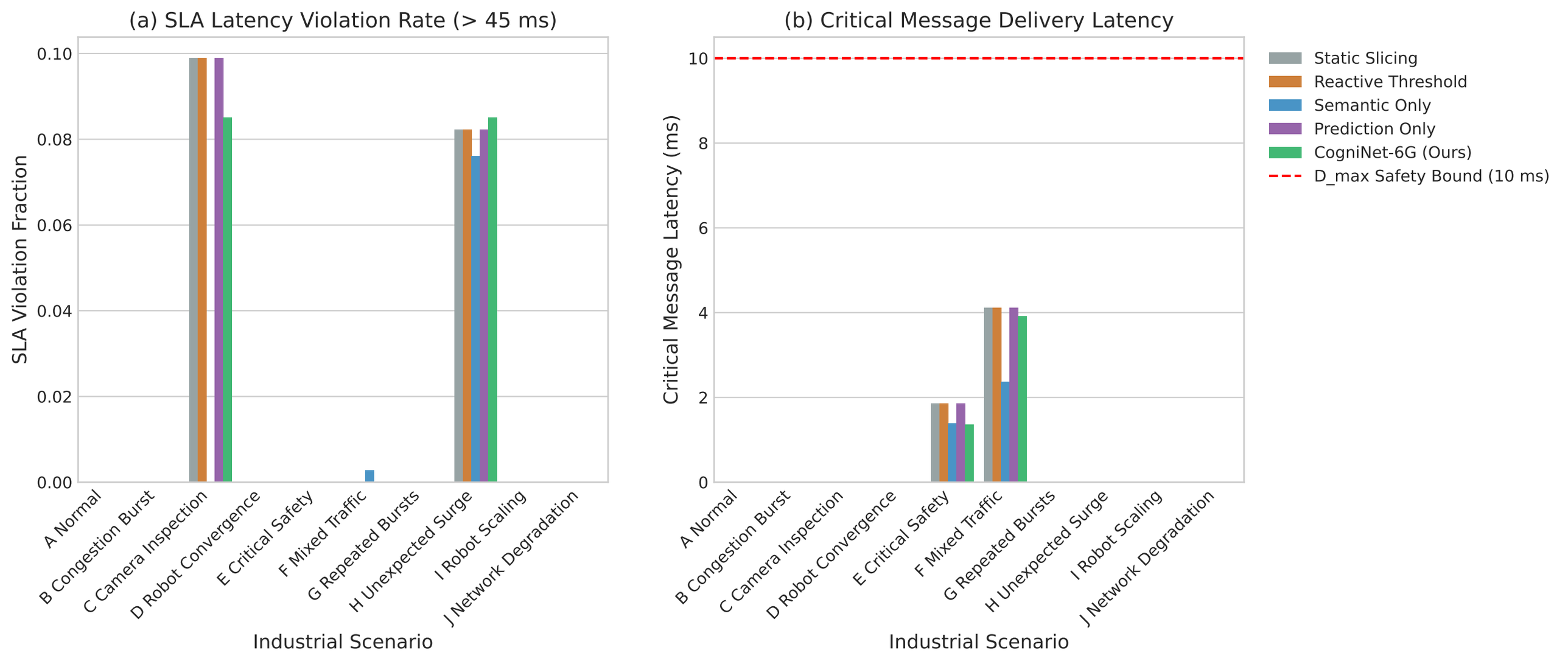
(a) SLA Latency Violation Rate (> 45 ms)
SLA Violation Fraction
0.00
0.02
0.04
0.06
0.08
0.10
A Normal
B Congestion Burst
C Camera Inspection
D Robot Convergence
E Critical Safety
F Mixed Traffic
G Repeated Bursts
H Unexpected Surge
I Robot Scaling
J Network Degradation
Industrial Scenario
(b) Critical Message Delivery Latency
Critical Message Latency (ms)
0
2
4
6
8
10
Static Slicing
Reactive Threshold
Semantic Only
Prediction Only
CogniNet-6G (Ours)
D_max Safety Bound (10 ms)
Industrial Scenario